

Erdosteine in children and adults with bronchiectasis (BETTER trial): study protocol for a multicentre, double-blind, randomised controlled trial.

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Bronchiectasis is a worldwide chronic lung disorder where exacerbations are common. It affects people of all ages, but especially Indigenous populations in high-income nations. Despite being a major contributor to chronic lung disease, there are no licensed therapies for bronchiectasis and there remain relatively few randomised controlled trials (RCTs) conducted in children and adults. Our RCT will address some of these unmet needs by evaluating whether the novel mucoactive agent, erdosteine, has a therapeutic role in children and adults with bronchiectasis. Our primary aim is to determine in children and adults aged 2-49 years with bronchiectasis whether regular erdosteine over a 12-month period reduces acute respiratory exacerbations compared with placebo. Our primary hypothesis is that people with bronchiectasis who regularly use erdosteine will have fewer exacerbations than those receiving placebo. Our secondary aims are to determine the effect of the trial medications on quality of life (QoL) and other clinical outcomes (exacerbation duration, time-to-next exacerbation, hospitalisations, lung function, adverse events). We will also assess the cost-effectiveness of the intervention. We are undertaking an international multicentre, double-blind, placebo-RCT to evaluate whether 12 months of erdosteine is beneficial for children and adults with bronchiectasis. We will recruit 194 children and adults with bronchiectasis to a parallel, superiority RCT at eight sites across Australia, Malaysia and Philippines. Our primary endpoint is the rate of exacerbations over 12 months. Our main secondary outcomes are QoL, exacerbation duration, time-to-next exacerbation, hospitalisations and lung function. The Human Research Ethics Committees (HREC) of Children's Health Queensland (for all Australian sites), University of Malaya Medical Centre (Malaysia) and St. Luke's Medical Centre (Philippines) approved the study. We will publish the results and share the outcomes with the academic and medical community, funding and relevant patient organisations. ACTRN12621000315819.

Airway "Resistotypes" and Clinical Outcomes in Bronchiectasis.

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Rationale: Chronic infection and inflammation shapes the airway microbiome in bronchiectasis. Utilizing whole-genome shotgun metagenomics to analyze the airway resistome provides insight into interplay between microbes, resistance genes, and clinical outcomes. **Objectives:** To apply whole-genome shotgun metagenomics to the airway microbiome in bronchiectasis to highlight a diverse pool of antimicrobial resistance genes: the "resistome," the clinical significance of which remains unclear. **Methods:** Individuals with bronchiectasis were prospectively recruited into cross-sectional and longitudinal cohorts (n = 280), including the international multicenter cross-sectional Cohort of Asian and Matched European Bronchiectasis 2 (CAMEB 2) study (n = 251) and two independent cohorts, one describing patients experiencing acute exacerbation and a further cohort of patients undergoing *Pseudomonas aeruginosa* eradication treatment. Sputum was subjected to metagenomic sequencing, and the bronchiectasis resistome was evaluated in association with clinical outcomes and underlying host microbiomes. **Measurements and Main Results:** The bronchiectasis resistome features a unique resistance gene profile and increased counts of aminoglycoside, bicyclomycin, phenicol, triclosan, and multidrug resistance genes. Longitudinally, it exhibits within-patient stability over time and during exacerbations despite between-patient heterogeneity. Proportional differences in baseline resistome profiles, including increased macrolide and multidrug resistance genes, associate with shorter intervals to the next exacerbation, whereas distinct resistome archetypes associate with frequent exacerbations, poorer lung

function, geographic origin, and the host microbiome. Unsupervised analysis of resistome profiles identified two clinically relevant "resistotypes," RT1 and RT2, the latter characterized by poor clinical outcomes, increased multidrug resistance, and *P. aeruginosa*. Successful targeted eradication in *P. aeruginosa*-colonized individuals mediated reversion from RT2 to RT1, a more clinically favorable resistome profile demonstrating reduced resistance gene diversity. Conclusions: The bronchiectasis resistome associates with clinical outcomes, geographic origin, and the underlying host microbiome. Bronchiectasis resistotypes link to clinical disease and are modifiable through targeted antimicrobial therapy.

[Bronchiectasis].

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Bronchiectasis is defined as irreversible dilatation of the bronchial lumen. Bronchiectasis not due to cystic fibrosis is the third most common chronic inflammatory disease of the airway, after asthma and chronic obstructive pulmonary disease (COPD). The pathogenesis of the disease is characterised by chronic dilatation, with irreversible and usually progressive destruction of the bronchial wall as a consequence of the vicious pathogenic circle described by E. Cole, of infection, inflammation, injury to the mucociliary system, and cyclical repair of the airway. The disease can have different causes, and usually presents with cough and chronic expectoration, and repeated, acute, infectious exacerbations. Suspected diagnosis is clinical, and must be confirmed by high resolution computerised tomography. The treatment comprises diverse regimens aimed at controlling infection and bronchial inflammation. Nutritional aspects should be considered as part of treatment, as well as the management of secretions, muscle training and the management of complications and comorbidities.

Bronchiectasis, Chronic Suppurative Lung Disease and Protracted Bacterial Bronchitis.

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Bronchiectasis is a structural airway disease characterized by dilated bronchi and bronchioles due to severe or recurrent lower airways inflammation. Bronchiectasis can occur as a result of chronic pulmonary aspiration. Bronchiectasis may also be associated with a wide variety of systemic diseases, which should be considered in the differential diagnosis. Children with bronchiectasis typically have a chronic or recurrent productive cough and carry a significant burden of disease with a considerable impact on quality of life. The diagnosis of bronchiectasis is made by high-resolution chest computerized tomography. Aggressive management of bronchiectasis is necessary to reduce the daily symptom burden and frequency of exacerbations. Chronic suppurative lung disease may be a precursor to bronchiectasis, only lacking the defining radiographic features of bronchiectasis. Children with chronic suppurative lung disease may have the same symptoms as children with bronchiectasis and should be treated similarly. Protracted bacterial bronchitis is defined as a cough lasting at least four weeks that responds to antibiotic therapy. Protracted bacterial bronchitis may occur following a viral respiratory tract infection. Protracted bacterial bronchitis can be treated with a prolonged course of empiric antibiotics. Further evaluation is necessary if a child with suspected protracted bacterial bronchitis does not adequately respond to antibiotics as chronic suppurative lung disease or bronchiectasis must be considered.

Non-cystic fibrosis bronchiectasis.

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Bronchiectasis is a chronic debilitating condition. Pathologically, a vicious cycle of infection and inflammation exists in the permanently damaged airways with patients suffering a persistent cough, chronic daily sputum production and recurrent chest infections. Once termed an 'orphan disease', the prevalence of bronchiectasis has become increasingly recognised over the past few decades. The associated burden of disease in terms of respiratory morbidity, effect on patients' health-related quality of life and the economic cost of long term management is significant and it has become apparent that more research into its causes and management is urgently needed. This article reviews what is currently known about bronchiectasis, its pathophysiology, aetiology and management strategies.

Bronchiectasis in children.

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Bronchiectasis is a chronic disease of the conducting airways that produces persistent productive cough, recurrent respiratory infectious exacerbations, and obstructive lung disease in children and adults. This article focuses on the grading and recommendations for chronic therapies of bronchiectasis caused by cystic fibrosis (CF)- and non-CF-related conditions. The scope of this article is to focus on outpatient treatment and not include as-needed treatment for mild or severe pulmonary exacerbations associated with bronchiectasis.

Chronic cough due to bronchiectasis: ACCP evidence-based clinical practice guidelines.

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Bronchiectasis is a condition that is characterized by the permanent dilation of bronchi with destruction of elastic and muscular components of their walls, usually due to acute or chronic infection. The cardinal symptom is a chronic productive cough. Review of articles cited in the systematic literature search, along with others found in Ovid MEDLINE and the Cochrane Library (including the Cochrane Database of Systematic Reviews, the Cochrane Controlled Trial Register, and the Database of Abstracts of Reviews of Effectiveness) from 1966 through 2003. High-resolution CT scanning of the chest is the preferred means of establishing the diagnosis of bronchiectasis. With the increasing use of antibiotics in the treatment of childhood infection in the last several decades, an increasing percentage of patients with bronchiectasis now have an underlying disorder that predisposes them to chronic or recurrent infection. These include cystic fibrosis, common variable immunodeficiency, HIV infection, primary ciliary dyskinesia, allergic bronchopulmonary aspergillosis, and chronic *Mycobacterium avium* complex infection. A variety of agents have been used to improve cough effectiveness and prevent infectious exacerbations in patients with bronchiectasis, with variable results. Chest physiotherapy offers a modest benefit in increasing sputum volume, but its long-term effectiveness is unknown. Selected patients with localized idiopathic bronchiectasis that causes intolerable symptoms despite maximal medical therapy should be offered treatment with surgery. Patients with exacerbations of bronchiectasis should be given antibiotics, with the choice of agents depending on the likely causative pathogens.

Pseudomonas aeruginosa in bronchiectasis: infection, inflammation, and therapies.

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Introduction: Bronchiectasis is a chronic endobronchial suppurative disease characterized by irreversibly dilated bronchi damaged by repeated polymicrobial infections and predominantly, neutrophilic airway inflammation. Some consider bronchiectasis a syndromic consequence of several different causes whilst others view it as an individual disease entity. In most patients, identifying an underlying cause remains challenging. The acquisition and colonization of affected airways by *Pseudomonas aeruginosa* represent a critical and adverse clinical consequence for its progression and management. **Areas covered:** In this review, we outline clinical and pre-clinical peer-reviewed research published in the last 5 years, focusing on the pathogenesis of bronchiectasis and the role of *P. aeruginosa* and its virulence in shaping host inflammatory and immune responses in the airway. We further detail its role in airway infection, the lung microbiome, and address therapeutic options in bronchiectasis. **Expert opinion:** *P. aeruginosa* represents a key pulmonary pathogen in bronchiectasis that causes acute and/or chronic airway infection. Eradication can prevent adverse clinical consequence and/or disease progression. Novel therapeutic strategies are emerging and include combination-based approaches. Addressing airway infection caused by *P. aeruginosa* in bronchiectasis is necessary to prevent airway damage, loss of lung function and exacerbations, all of which contribute to adverse clinical outcome.

Pathophysiology of Chronic Bronchial Infection in Bronchiectasis.

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Bronchiectasis is a complex and heterogeneous disease. Its pathophysiology is poorly understood, but chronic bronchial infection plays an important role in its natural history, and is associated with poor quality of life, more exacerbations and increased mortality. *Pseudomonas aeruginosa*, *Haemophilus influenzae* and *Staphylococcus aureus* are the most common bacteria related to chronic bronchial infection. Non-tuberculous mycobacteria, fungi and respiratory viruses are also present during clinical stability, and may increase the risk of acute exacerbation. Chronic inflammation is present in bronchiectasis, especially neutrophilic inflammation. However, macrophages and eosinophils also play a key role in the disease. Finally, airway epithelium has innate mechanisms such as mucociliary clearance and antibacterial molecules like mucins and antimicrobial peptides that protect the airways from pathogens. This review addresses how the persistence of microorganisms in the airways and the imbalance of the immune system contribute to the development of chronic bronchial infection in bronchiectasis.

Dual bronchodilators in Bronchiectasis study (DIBS): protocol for a pragmatic, multicentre, placebo-controlled, three-arm, double-blinded, randomised controlled trial studying bronchodilators in preventing exacerbations of bronchiectasis.

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Bronchiectasis is a long-term lung condition, with dilated bronchi, chronic inflammation, chronic infection

and acute exacerbations. Recurrent exacerbations are associated with poorer clinical outcomes such as increased severity of lung disease, further exacerbations, hospitalisations, reduced quality of life and increased risk of death. Despite an increasing prevalence of bronchiectasis, there is a critical lack of high-quality studies into the disease and no treatments specifically approved for its treatment. This trial aims to establish whether inhaled dual bronchodilators (long acting beta agonist (LABA) and long acting muscarinic antagonist (LAMA)) taken as either a stand-alone therapy or in combination with inhaled corticosteroid (ICS) reduce the number of exacerbations of bronchiectasis requiring treatment with antibiotics during a 12 month treatment period. This is a multicentre, pragmatic, double-blind, randomised controlled trial, incorporating an internal pilot and embedded economic evaluation. 600 adult patients (≥ 18 years) with CT confirmed bronchiectasis will be recruited and randomised to either inhaled dual therapy (LABA+LAMA), triple therapy (LABA+LAMA+ICS) or matched placebo, in a 2:2:1 ratio (respectively). The primary outcome is the number of protocol defined exacerbations requiring treatment with antibiotics during the 12 month treatment period. Favourable ethical opinion was received from the North East-Newcastle and North Tyneside 2 Research Ethics Committee (reference: 21/NE/0020). Results will be disseminated in peer-reviewed publications, at national and international conferences, in the NIHR Health Technology Assessments journal and to participants and the public (using lay language). ISRCTN15988757.
